

Shenandoah National Park

National Park Service

Meadow Run Watershed Liming

Statement of Work

STATEMENT OF NEED

The National Park Service (NPS) is anticipating award of a single service contract to apply approximately 5,250 tons of limestone sand or equally efficacious material (aka “liming”) across the Meadow Run Watershed (aka “project area”), a largely remote and inaccessible backcountry area of Shenandoah National Park (aka “the Park”). This area is difficult to access on foot; trails roughly run the perimeter of the project area. Access is impossible for wheeled vehicles as there are no roads in the project area, the majority of which is congressionally designated Wilderness.

PERFORMANCE PERIOD

CLIN 0010 – Base Period: Liming Operations Winter 2025-2026

August 2025 through June 2026 (Liming Operations 01 December 2025 through 28 February 2026)

CLIN 0020 – Increase Quantity Option: Additional Acreage Winter 2025-2026

August 2025 through June 2026 (Liming Operations 01 December 2025 through 28 February 2026)

CLIN 0030 – Option Period One: Liming Operations Winter 2026-2027

September 2026 through June 2027 (Liming Operations 01 December 2026 through 28 February 2027)

CLIN 0040 – Increase Quantity Option: Additional Acreage Winter 2026-2027

September 2026 through June 2027 (Liming Operations 01 December 2026 through 28 February 2027)

Work in Park can commence upon all pre site work deliverables being received and approved and during the time windows stipulated below for access to staging areas (2.1.2.1) and liming operations (2.2.1).

The liming operations will be conducted only during daylight hours, Monday through Friday. Weekend work may be approved based on availability of park oversight. A 96-hour advance notice is required for all liming operations, (2.2.1 and 2.3.2). Contractor will not deviate from the Monday through Friday without receiving written concurrence from the Contracting Officer (CO) or Contracting Officer’s Representative (COR).

BACKGROUND

Meadow Run is a stream located in the South District of Shenandoah National Park (the park). The stream’s watershed is underlain by silica-rich bedrock. Soils that form on this type of bedrock have

limited capacity to neutralize acidic deposition. The resulting acidic groundwater within these soils eventually flows into the tributaries of Meadow Run, affecting its water quality. Consequently, some fish species, including brook trout, many insect species, and other aquatic organisms struggle to survive in Meadow Run. Because low pH is affecting Meadow Run's aquatic life, the Virginia Department of Environmental Quality (VADEQ) formally lists the stream as impaired waters under Section 303(d) of the Clean Water Act. Acidification also adversely affects songbird, plant, and forest health. Degraded ecosystem health in turn diminishes the visitor experience.

Meadow Run is the most acidified stream in the park, has a 30-year history of scientific research on acid rain effects and is the ideal size for a restoration effort. The ecosystem restoration project area covers 2,150 acres of the park's 199,234 acres, including approximately 1,700 acres of congressionally designated wilderness. Because of the watershed's underlying bedrock geology, the soil and stream cannot recover naturally to pre-pollution conditions for well over 100 years in the current climate. The effects of climate change, with potentially higher average temperatures and more extreme storms is expected to exert additional acidification impacts over time, increasing leaching and loss of soil elements, driving higher stream flows, and producing more intense episodic acidification events. To restore the ecosystem function and improve its resilience to climate change, approximately 5,250 tons of limestone sand will be applied in the project area to replace elements lost from soils due to acidification and increase the acid neutralizing capacity (ANC) in the stream.

Improvements to the watershed should be observable within a year and are expected to have beneficial impacts on soils for well over 100 years. Liming will reduce soil acidity and toxic aluminum. The pH in the stream is expected to increase above 6.0, allowing Meadow Run to be removed from VADEQ's list for Impaired Water and thereby meeting requirement under federal law to develop and implement a mitigation plan.

The Environmental Assessment for this project was completed in January 2021. A Finding of No Significant Impact was issued in April 2021.

ATTACHMENTS

- 1 – Meadow Run Watershed Restoration Project Environmental Assessment
- 2 – Meadow Run Watershed Restoration Project Finding of No Significant Impact
- 3 – Meadow Run Shapefiles [packaged in SHEN_MR_Solicitation.zip]

OVERALL PROJECT ACTIVITIES

1. The NPS will provide limited staging along Skyline Drive. The contractor is responsible for acquiring any additional off-park staging needed to complete operations and manage project risk.
2. The NPS will make all necessary park closures for the project area, at staging areas, and along Skyline Drive as coordinated between the NPS and the contractor. The contractor is responsible for safety and controlling access immediately around operations in and around staging areas.

3. The NPS will parcel the project area, set the dosage for each parcel, and prioritize them for treatment.
4. The contractor is responsible for acquiring limestone sands, calibrating treatment method, managing logistics, completing the treatment, providing treatment documentation to the NPS, and returning staging areas to pre-operation conditions.
5. The contractor will turn in all project documentation and a final report.

PRICING

The pricing schedule is entailed in the Instructions to Offerors and will be incorporated in the contract.

Base and Option Periods are to be provided in a lump sum unit for the services to be covered. Increase Quantity Options for both the Base and Option Periods are broken down by liming dose and year of service. The total quantity of acres limed over the duration of the contract cannot exceed the total acreage of all parcels.

The Government intends, based on funding availability, to unilaterally include Additional Acreage under the Increase Quantity Options at the time of award for both the Base Period and, should it be exercised, the Option Period.

The Government may also, based on funding availability, be able to include Additional Acreage in an Increase Quantity Option during an ongoing period of performance. The further Additional Acreage will be agreed on bilaterally. To do so, the Government will contact the Contractor to establish feasibility of including the Additional Acreage with the ongoing effort. The contractual price list will not be subject to change based on such a request and will stand firm for the duration of this contract. In the event it is determined that including Additional Acreage under the Increase Quantity Option is not feasible in the Base Period, there is no guarantee of exercising the Option Period.

PAYMENT SCHEDULE

Due to the magnitude and logistical complexity of this project, the Government authorizes the following payment schedule for the Base Period and potential Option Period. The Contractor may also choose to combine invoicing or bill one lump sum at the conclusion of the project for the Base Period or Option Period, respectively.

Note: 'Base effort' in the table below is the total area to be covered for the Base Period or Option Period.

Invoicing Schedule:

Description	Precent	Invoice Requirement
Work Plan Completion and Pre-Operations Staging	10%	Work Plan has been accepted by the Government and no earlier than first day of Liming Operations
Liming Operations 50% Completion	40%	Supporting documentation can show at least half of the base effort limestone application has been completed and has been accepted by the Government.
Liming Operations 100% Completion	40%	Supporting documentation can show the remainder of the base effort has been completed and has been accepted by the Government.
Final Inspection and Documentation Completion	10%	Final Inspection complete; Documentation and Final Report accepted by the Government.

Increase Quantity Option Additional Acreage may be invoiced after the period's base operations are 100% complete and no more often than once every two weeks. The contractor may invoice for all additional acreage with supporting documentation showing the effort has been completed and has been accepted by the Government. The invoice will list each parcel or sub-parcel completed and round the acreage to the nearest tenth (0.1) of an acre, as defined (see Table 1, below).

For example:

Additional Acreage – Partial liming of parcel C06 at 2.25 tons = \$1.00 x 2.6 acres = \$2.60

DATA RIGHTS

The Government asserts all data and deliverables are provided with unlimited data rights. These data are of the type that the agency itself distributes to the public and should be disseminated without restriction (FAR 27.404-3 (a)(2)(iii) and (v)). The contractor is not prohibited for utilizing the same data.

STATEMENT OF WORK

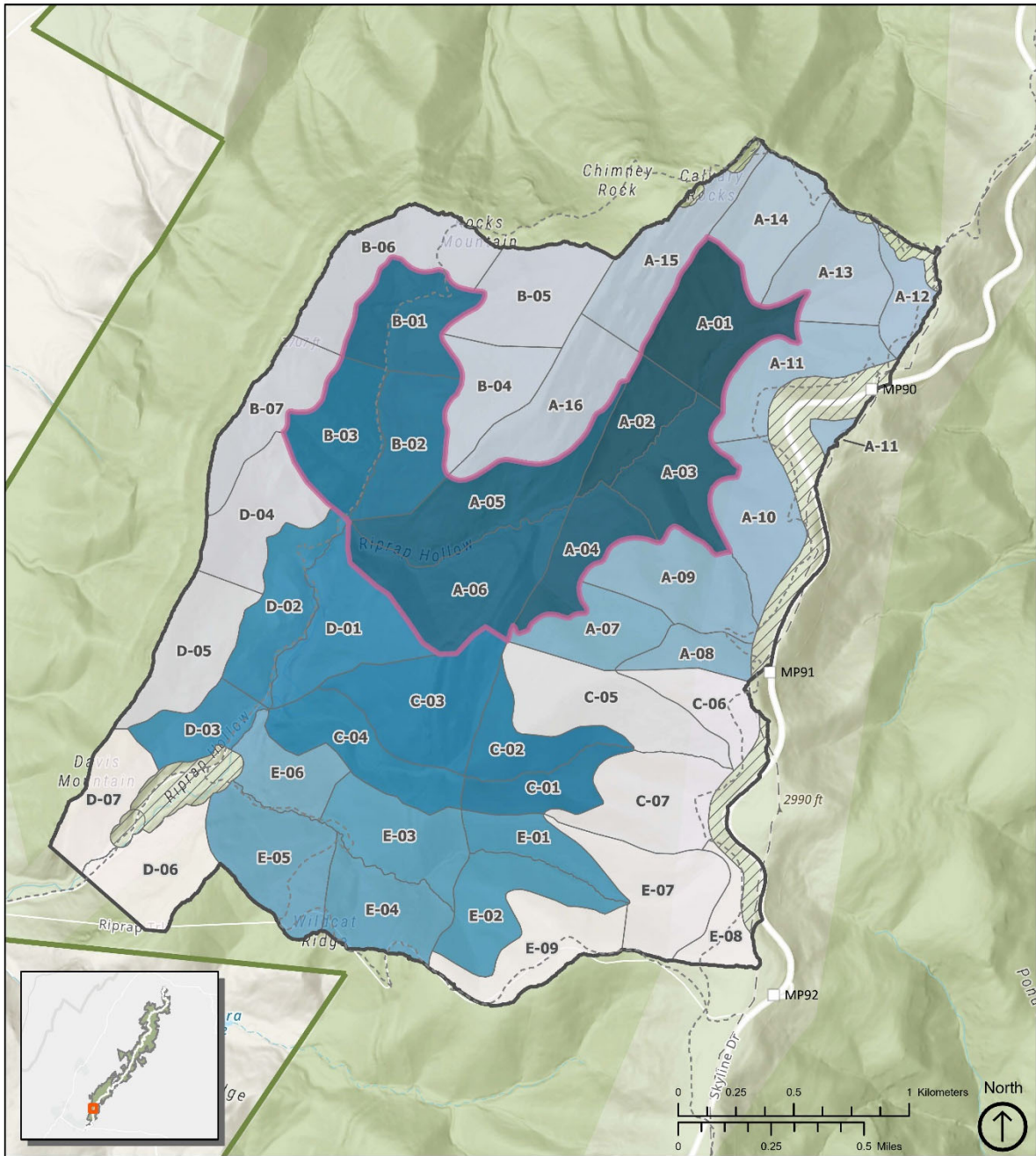
Definition of Meadow Run Project Area

The Meadow Run Project Area has been broken into 46 parcels as defined by the shapefile SHEN_MR_Parcels.shp accompanying this Statement of Work; shown in the map below, Figure 1; and listed in Table 1, on the following pages, along with the initial prioritized liming order, parcel size in acres, and target liming dose. Areas within the project area that will be excluded from liming (aka "exclusion zones") are also identified in the map below and defined by the shapefile SHEN_MR_ExclusionZone.shp accompanying this Statement of Work. The NPS may adjust the prioritization of the parcels and split parcels as needed to administer this contract.

Table 1: Meadow Run Liming Parcels

Name	Liming Order	Acres	Dose (tons/acre)
A-01	1	60.6	3.0
A-02	2	30.8	3.0
A-03	3	47.3	3.0
A-04	4	36.2	3.0
A-05	5	62.9	2.25
A-06	6	71.5	2.25
B-01	7	48.4	2.25
B-02	8	45.2	2.25
B-03	9	46.7	2.25
C-01	10	28.2	3.0
C-02	11	31.4	3.0
C-03	12	65.1	2.25
C-04	13	45.2	2.25
D-01	14	56.1	2.25
D-02	15	45.1	2.25
D-03	16	27.9	2.25
E-01	17	26.1	3.0
E-02	18	45.8	3.0
E-03	19	48.2	2.25
E-04	20	44.2	2.25
E-05	21	57.6	2.25
E-06	22	31.2	2.25
A-07	23	35.9	3.0

Name	Liming Order	Acres	Dose (tons/acre)
A-08	24	19.7	3.0
A-09	25	52.4	3.0
A-10	26	47.5	3.0
A-11	27	49.3	3.0
A-12	28	18.3	3.0
A-13	29	46.8	3.0
A-14	30	43.4	3.0
A-15	31	49.8	2.25
A-16	32	54.3	2.25
B-04	33	37.8	2.25
B-05	34	52.6	2.25
B-06	35	57.2	2.25
B-07	36	39.6	2.25
D-04	37	47.2	2.25
D-05	38	46.9	2.25
C-05	39	62.2	3.0
C-06	40	32.5	3.0
C-07	41	48.5	3.0
E-07	42	51.5	3.0
E-08	43	19.2	3.0
E-09	44	46.1	3.0
D-06	45	44.2	2.25
D-07	46	33.9	2.25



Base Activity

Meadow Run Watershed Restoration

Produced by the National Park Service
Shenandoah National Park
July 2024
Map ID: 20230620_3



Figure 1: Meadow Run Liming Parcels and Exclusion Zones

A. Base Period: Liming Operations Winter 2025-2026 (CLIN 0010)

This activity is to lime parcels A-01 through A-06 and B-01 through B-03 of the Project Area, as defined above.

Liming operations in Meadow Run will be in line with the Meadow Run Environmental Assessment as stipulated in the Finding of No Significant Impact. The contractor shall comply with and meet all relevant requirements under all applicable laws and regulation, including the Superintendent's Compendium, while executing the liming operations.

1.0 Post Award Project Meeting and Work Plan

1.1 Post Award Project Meeting and Site Visit

Contractor will attend a Post Award Meeting. Contractor and all principal team members will meet with NPS representatives to clarify tasks; review the Contractor's proposed Work Plan; and thoroughly discuss safety planning, logistics, and schedule. An in-person Site Visit is required to orient the Contractor to the liming site and staging areas along Skyline Drive, and the NPS to any off-park staging. Both activities may be conducted on consecutive days.

1.2 Work Plan

Based on the outcome of the Post Award Project Meeting and Site Visit, the Contractor shall refine their proposal and submit a Draft Work Plan for review by NPS representatives within 21 calendar days of completing the Post Award Meeting. After NPS review, the Contractor shall incorporate all NPS comments and edits for acceptance by NPS representatives within 7 calendar days of review. Refinements will remain within scope of this Statement of Work (SOW) and should not alter the contract award value. The Work Plan shall include:

1. An Executive Summary providing an overview of the proposed liming approach, and highlighting: innovative aspects used, assumptions made in preparation of the Work Plan, and critical risks beyond the Contractor's ability to mitigate.
2. A detailed Description of the Liming Process, including:
 - a. The sequence of steps or events for the project
 - b. The personnel positions required to execute the work plan
 - c. The equipment required to execute the work plan, and how it will be configured and maintained
 - d. The liming product to be applied to the Project Area including the guaranteed screen analysis from the quarry(ies)
 - e. How the contractor will communicate with the NPS and internally during active liming operations, at a minimum: timing, method, and equipment
 - f. How the contractor will address requirements outlined here in the SOW
3. A detailed Treatment Verification Plan addressing requirements outlined in the SOW and describing:
 - a. How the liming equipment will be calibrated, operated, and monitored for correct liming dose

- b. How the liming application will be recorded and verified for each parcel
 - c. The data collected to support a and b, above, including format, metadata, sources and magnitude of measurement error (accuracy and precision), and availability/delivery timeline. This data will include geospatial data showing, at a minimum, when, where, and how much limestone sand was dispersed.
- 4. A detailed Safety Plan describing how their work will be conducted in compliance with all local, state, and federal required safety standards. In particular, the plan will also address:
 - a. How the Contractor will manage controlling access immediately around operations around staging areas and steps to take in the event people do enter the site, both planned (ex: scheduled NPS project staff and deliveries from subcontractors) and unplanned (ex: public entering the closure area)
 - b. Go/No-Go criteria for conducting liming operations
 - c. Equipment refueling
 - d. Site modifications needed in staging areas for safe liming operations (ex: needed vegetation removal or special equipment temporarily installed on site)
 - e. Steps to take in the event of a mishap or accident.
- 5. A detailed Project Schedule showing a target timeline for executing the Work Plan and liming all parcels in the Project Area (both the Base Period activity and all remaining acreage in the Increase Quantity Option: Liming Additional Acreage) taking into account inclement weather days, Skyline Drive closures due to winter storms with subsequent clearing, and otherwise manages risk of project completion.
- 6. A Risk Management Plan identifying, assessing, and outlining any mitigation for performance, schedule, safety, and any other project risks.

2.0 Liming Operations

Liming Operations are the core of this project. Managing the project logistics and systematic application on the landscape, both in a safe manner and in accordance with this SOW, including the Environmental Assessment and Finding of No Significant Impact, is of paramount importance. The Contracting Officer or Contracting Officer's Representative can stop work at any time for safety of the public or protection of the resource.

2.1 Logistics

The Contractor will adhere to the following logistical considerations for planning and during operations:

2.1.1 Project Area

The project area is difficult to access on foot. Three trails, in part, are located within the project area: Riprap Trail, Wildcat Ridge Trail, and the Appalachian Trail. There are no roads in the project area. Much of the project area consists of steep terrain and Congressionally designated Wilderness. Wheeled vehicles are prohibited within the project area. Building new roads, trails,

or other infrastructure is prohibited. Improving existing trails is prohibited. Ground disturbance in the project area is prohibited without explicit written permission from the CO or COR.

The NPS will close the Project Area to all public visitor use for the duration of the Liming Operations.

2.1.2 Staging Areas

The Government will provide up to four staging areas for use by the contractor along Skyline Drive. Liming material may be piled in up to three of these staging areas. The Contractor will choose from the areas listed below in Table 2 and annotated on the map, Figure 2. The size of each location shown reflects site restrictions in this SOW. The contractor is not required to use these areas exclusively or at all. Off-park staging should be located in the polygon to the west of the project area as annotated on the map. The contractor is responsible for securing any off-park staging.

Table 2: Possible Staging Areas Along Skyline Drive

Location Name	Mile Marker	Vista Aspect	Size (sq ft)	Additional Site Restrictions
Trayfoot Mountain Overlook	86.8	East	6,530	None
Riprap Overlook	91.4	West	9,680	Sand and Equipment prohibited on grass in median
Moormans River Overlook	91.9	East	7,490	None
Turk Mountain Overlook	93.6	West	10,830	Sand and Equipment prohibited on grass in median
Sawmill Run Overlook	95.3	West	11,240	Sand and Equipment prohibited on grass in median
Beagle Gap	99.9	West	6,520	Sand and Equipment prohibited on grass surrounding gravel

2.1.2.1 Access

The Contractor will have access to the desired staging areas along Skyline Drive from 03 November 2025 through 10 April 2026 or passing final inspection, whichever is earlier. In the event of winter storms and any associated storm damage, Skyline Drive will be closed, and the Contractor will not have access to these areas until the NPS has completed road clearing operations.

The NPS will close staging areas to all public visitor use for the duration the Contractor has access to the staging areas. The NPS will close portions of Skyline Drive to all use, as needed, to accommodate liming operations. The contractor is responsible for safety and controlling access immediately around operations in and around staging areas.

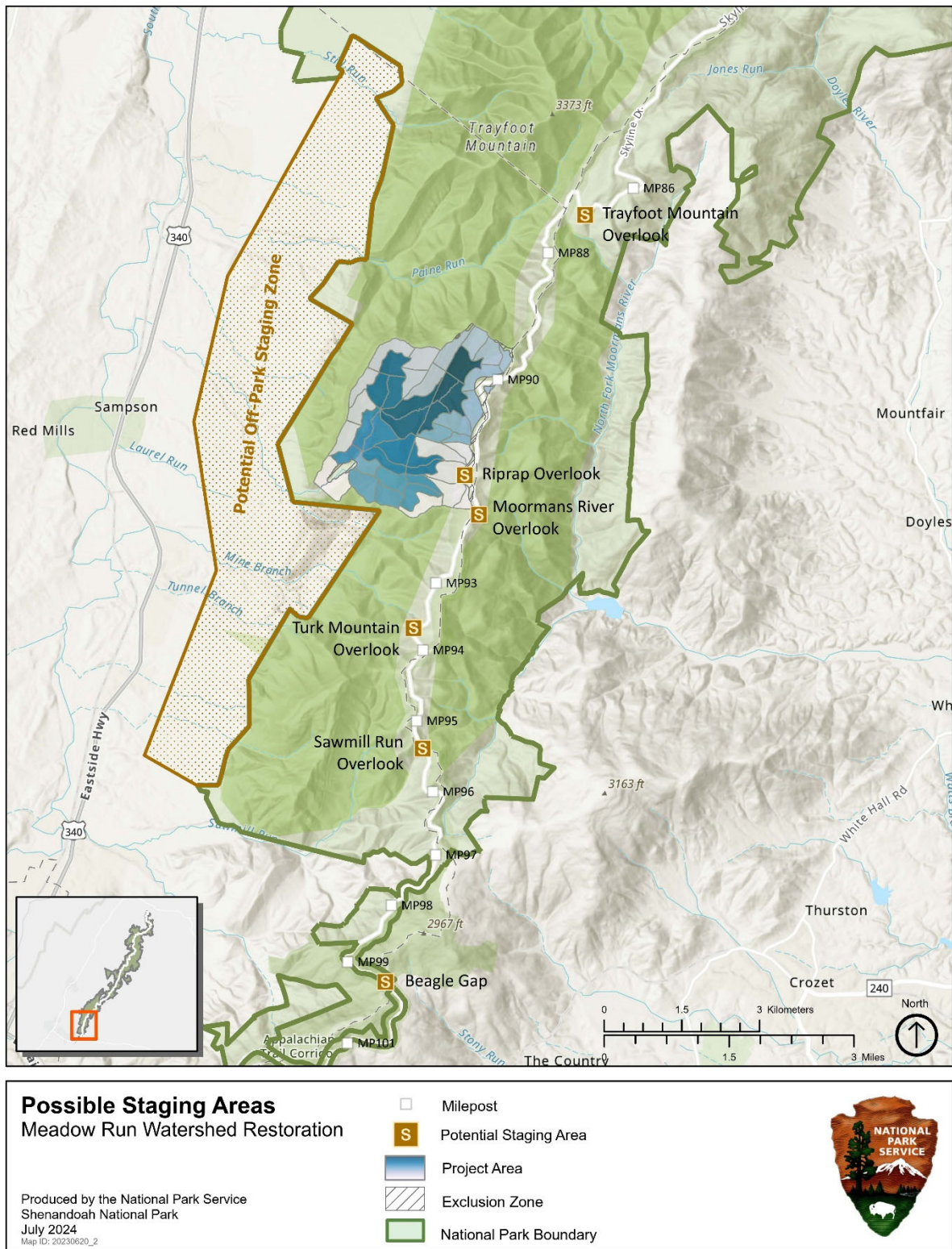


Figure 2: Meadow Run Possible Staging Areas

The NPS will periodically access staging areas, both on and off the park, to meet with the contractor, provide requisite support for the project, and ensure compliance with this Statement of Work.

2.1.2.2 Site Management

The contractor will take measures to protect pavement and soil from the limestone sand in and around all staging areas. A waterproof barrier and construction matting will be placed for protection of the asphalt during operations. Limestone sand can be stockpiled on the construction matting and will be covered with a waterproof barrier. The stockpiles will be surrounded with erosion control.

The contractor will comply with and meet all relevant requirements under the Clean Water Act, including management of stormwater-related non-point source pollutants under the National Pollutant Discharge Elimination System. The contractor will implement best management practices for drainage and sediment erosion control at staging areas to prevent or reduce non-point source runoff and minimize soil loss from the limestone sand stockpile and any unpaved surfaces. The contractor will inspect and clean out erosion control structures periodically and remove them only after staging areas are fully restored.

The contractor will follow a refueling and maintenance plan designed to prevent potential contamination of surface waters and groundwater caused by accidental spills for all equipment and vehicles. The refueling and maintenance plan will be identified in the Work Plan.

Steel tracked vehicles and equipment are prohibited in park staging areas and on Skyline Drive. All limestone sand and equipment will stay a minimum of 12' from the stone walls and curbs at all times. Limestone sand is prohibited on the surface of Skyline Drive.

The contractor may use Skyline Drive to maneuver equipment and vehicles (for example, loading and unloading trailered equipment). However, Skyline Drive must remain passable for emergency traffic at all times.

2.1.2.3 Vegetation

The contractor will not disturb any vegetation, alive or dead, within the Park unless otherwise allowed in this SOW. The government may remove vegetation at staging areas along Skyline Drive deemed a safety issue and requested in writing by the contractor. There is no guarantee of removal. Live trees will not be removed. The contractor may use mowed grassy areas immediately adjacent to the pavement to maneuver equipment and vehicles except where there is a noted prohibition in Table 2, above.

The contractor will take measures to prevent or minimize the establishment and spread of non-native invasive plants, noxious weeds, and the spread of diseases through the measures outlined in the Meadow Run Environmental Assessment. Staging areas and the land within 100 yards of the staging areas, both on and off the park, should be reasonably free of invasive plants and their seed. The NPS will treat the staging areas along Skyline drive before 31 October 2025. The

contractor is responsible for management of plants and seed on any off park staging. The government reserves the right to periodically inspect off park staging to ensure compliance with this SOW.

2.1.2.4 Food Storage, Refuse, and Sanitation

The Contractor shall comply with and meet all relevant requirements under all applicable laws and regulations, including the Superintendent's Compendium, regarding Food Storage, Refuse, and Sanitation. The Contractor must make their own arrangements to comply.

2.1.3 Limestone Product

2.1.3.1 Composition

The Contractor will use Dolomitic Limestone as a sand product, American Association of State Highway and Transportation Officials (AASHTO) #10 aggregate product, or another equally efficacious material, as identified in the approved Work Plan.

2.1.3.2 Delivery from the Quarry

The Contractor is responsible for limestone sand acquisition and delivery. Upon arrival, limestone sands will be stored as outlined above.

The contractor will adhere to the maximum loads on Skyline Drive as stipulated by the Superintendent's Compendium. The CO/COR is the point of contact for any permitting. The Contractor will submit proof of purchase and delivery as part of the Final Report and Documentation Package.

2.2 Operations

The Contractor will adhere to the Environmental Assessment, the Finding of No Significant Impact, and the following considerations for planning and during operations:

2.2.1 Dates

Liming Operations shall be conducted during daylight hours, as defined by the US Naval Observatory, on weekdays, Monday through Friday, from 01 December 2025 through 28 February 2026. Operations will not occur on government holidays unless approved by the government. Weekend operations may be approved by the government should it become necessary. Sunday and holiday operations will be conducted exclusively within the park's boundaries and will necessitate the use of staging along Skyline Drive. All operations must be scheduled at least 96 hours in advance with the COR. Contractor shall not conduct weekend or holiday operations without receiving written concurrence from the COR.

2.2.2 Limestone Application

The Contractor will lime each parcel to the dose identified in Table 1. The total applied limestone sands will be +5 tons/-1 ton of the total limestone sand required to lime each parcel based on its acreage, also in Table 1, measured and calculated per parcel. Care should be taken to manage compounding measurement errors.

Lime will be applied in a systematic and uniform manner across each parcel following the approved Work Plan. The Contractor will pay special attention to airborne drift and make an effort to minimize it during application. Applying limestone sands outside of the identified parcels is prohibited, i.e. outside of the Project Area or within an identified Exclusion Zone.

The NPS will use the Work Plan and the data delivered to the NPS as outlined below in sections 1.3.7 Incremental Data Delivery and 2.0 Documentation and Final Report, including sub-sections, to help determine if the Contractor has met the requirements for Limestone Application.

2.3 Communications

The complexity of this project requires regular, recurring, and timely communication between the Contractor and NPS. The Contractor shall attend all meetings and submit all information as outlined below. The Contractor can plan to receive timely Skyline Drive status information from the NPS as outlined below. The Contractor will make arrangements to manage internal communications during the Liming Operations and contact the NPS in case of emergency as also outlined below.

2.3.1 Weekly Project Meeting

The Contractor shall meet with the NPS weekly to review project progress against the work plan, seek any clarification on any point, and relay any necessary information. The meeting may be held via virtual teleconference. Most meetings should not regularly last more than 1 hour. Weekly Project Meetings will commence the week following the Post Award Project meeting and continue through the acceptance of the Final Report.

2.3.2 Four-Day Outlook Notification

The Contractor will notify the COR in writing not less than 96 hours in advance of their intent to commence liming operations on a date so the NPS can make arrangements to manage necessary closures and post alerts. The contractor will notify the COR of their intent to suspend planned operations (for example, winter storms are forecasted or equipment needs maintenance) as soon as known so the NPS can forego or lift unnecessary closures.

The Four-Day Outlook may be contained in the Daily Planning Brief or Debrief on days the contractor is conducting liming operations. There are otherwise no restrictions as to date or time the contractor may send notification to the COR. The contractor may opt to file notice in advance of the 96 hours in lieu of doing so on weekends and holidays.

2.3.3 Status of Skyline Drive Notification

The NPS will provide notification to the Contractor of the status of Skyline Drive, confirming needed closures for liming operations and alerting them to closures and restricted use as a result of winter weather and road conditions. This notification may come from a mass messaging platform such as Everbridge.

The Contractor will adhere to Skyline Drive closures and use restrictions.

2.3.4 Daily Planning Brief

The Contractor will provide the COR with a Daily Planning Brief every day before liming operations begin. The Planning Brief will be a short, written communication and accompanied by an in-person meeting or virtual teleconference with the COR. The COR will have opportunity to seek clarification on any point and relay necessary information to the contractor. The meeting should last no more than twenty minutes. This brief will contain: planned start and stop times, parcels identified for liming, any logistics notes, and any safety messaging.

2.3.5 Active Liming Operations Communications

The Contractors may choose any standard communication methods that work for their operations. Cell coverage is not reliable throughout most areas of Shenandoah National Park, however, it may be available at overlooks around the Project Area and available staging areas depending on your carrier(s). In case of emergency, the contractor must be able to contact the Park Communications Center via two-way radio on the Park's frequencies; the Park's frequencies will only be used in case of emergency. These communication plans will be detailed in the Work Plan.

2.3.6 Daily Debrief

The Contractor will provide the COR with a Daily Debrief every day at the conclusion of liming operations. The Debrief will be a short, written communication and accompanied by an in-person meeting or virtual teleconference with the COR. The COR will have opportunity to seek clarification on any point and relay necessary information to the contractor. The meeting should last no more than twenty minutes. This brief will contain: the day's progress in liming the parcels, obstacles encountered, opportunities ahead, changes to plans for the next three days, and progress against the project schedule from the approved Work Plan.

2.3.7 Incremental Data Delivery

The Contractor will incrementally deliver the record of liming as outlined in the approved Work Plan by close of business (COB) the next workday after the liming is accomplished. This will consist of a geospatial file showing when, where, and how much limestone sand was dispersed and may consist of additional information such as load slips from limestone delivery or stand-alone loadcell data.

2.4 Restoration and Cleanup

Once liming operations are complete, the Contractor has 30 working days to complete restoration work, clean-up, and request an inspection. Restoration that requires weather conditions outside of this window can be accomplished and inspected separately at a later date with written permission from the CO or COR.

The Contractor will restore all staging areas along Skyline Drive, paved and unpaved, disturbed during project implementation to pre-implementation conditions. All plant species and materials (e.g., rolled erosion control products) used for revegetation must be approved by the NPS and any replanting and reseeding will be done within the optimal growing season. Any asphalt and

stonework restoration must be coordinated with the NPS. Restoration of staging areas off the park are subject to the contractor's negotiated terms of use.

The Contractor will complete the following cleaning operations before requesting final inspection:

1. Remove tools, construction equipment, machinery, and surplus material from Project site.
2. Clean staging areas and surrounding landscape of rubbish, waste material, litter, and other foreign substances.
3. Remove petrochemical spills, stains, and other foreign deposits.
4. Sweep paved areas broom clean. Rake grounds that are neither planted nor paved to smooth, even-textured surface, as directed by the NPS.

Final cleaning and waste removal operations will comply with local laws and ordinances and Federal and local environmental and antipollution regulations.

2.5 Final Inspection of Staging Areas

The NPS, upon written notification by the contractor that they are ready for a final inspection, will conduct a final inspection of the staging areas. The NPS will proceed with inspection on an agreed upon date upon receipt of written request.

If staging area restoration and cleanup is not determined complete following final inspection, the NPS will notify the Contractor in writing. The Contractor shall request a new final inspection after completing work. Payment for the site cleanup and final deliverables will be withheld until final inspection is complete.

3.0 Documentation and Final Report

The Contractor shall submit a Draft Final Report and Documentation Package for review by NPS representatives within 28 calendar days of completing the Final Inspection. After NPS review, the Contractor shall incorporate all NPS comments and edits for acceptance by NPS representatives within 14 calendar days of review.

The Final Report will include:

1. A description of the work completed
2. An analysis of the dose delivered in each parcel against the method outlined in the Treatment Verification Plan
3. A comparison of the Project Schedule and target timeline to the actual completion schedule
4. Challenges and risks encountered during the work and how they were managed
5. Any safety issues and incidents and how they were addressed
6. Recommendations for process improvement for future similar projects

The Documentation Package will include:

1. Evidence of limestone acquisition; this may include copies of all invoices and load slips from liming material
2. Data collected and created as part of the Treatment Verification Plan. This includes all equipment instrumentation and geospatial data, as well as a clean library containing all data included in the Incremental Data Delivery. All geospatial data will be in a commonly used file type, i.e. Esri shapefile or Google kml/kmz.
3. Still photography and videography of liming operation, including: 100 still images and 10 videos, each 2-10 minutes in length; a minimum of 25% of the content should be from the perspective of the application crew(s), a novel perspective of the application, and/or close up of their same work. Videography should be captured at 4K resolution or higher. Photography should be shot in RAW format at the highest resolution any camera allows.

Deliverables

- A1.1 Post Award Project Meeting
- A1.2 Site Visit
- A1.3 One (1) Draft Work Plan for review by NPS (format: Microsoft (MS) Word)
- A1.4 One (1) Final Work Plan that addresses and incorporates all NPS comments and edits of the Draft Work Plan (format: 508 compliant Adobe.pdf)
- A2.1 Limestone Sand Application for each Parcel
- A2.2 Weekly Project Meetings
- A2.3 Four-Day Outlook Notifications
- A2.4 Daily Planning Briefings
- A2.5 Daily Debriefs
- A2.6 Incremental Data Deliveries
- A2.7 Final Inspection of Staging Areas
- A3.1 One (1) Draft Final Report (format: MS Word)
- A3.2 One (1) Draft Documentation Package
- A3.3 One (1) Final Report that addresses and incorporates all NPS comments and edits of the Draft Final Report (format: 508 compliant Adobe .pdf)
- A3.4 One (1) Final Documentation Package that addresses and incorporates all NPS comments and edits of the Draft Documentation Package

B. Increase Quantity Option: Liming Additional Acreage Winter 2025-2026 (CLIN 0020)

The NPS may decide to lime additional acreage during the Base Period: Liming Operations Winter 2025-2026, as described above. This activity is to lime up to an additional not to exceed 1,590 acres of the Project Area. The NPS will identify which parcels or sub-parcels (if an entire parcel is not funded) will be limed. Each acre will be limed to the dose of the corresponding parcel where it is located, as described in Table 1.

Liming Additional Acreage for this Option will be completed in conjunction with the effort of Base Period: Liming Operations Winter 2025-2026, as described above. Liming Operations will be conducted in sequence with otherwise funded work and follow the same requirements as detailed above, respectively, and as described in the approved Work Plan. Inputs will be provided for the corresponding Final Report and Documentation Package. Specific changes to any requirements are below.

1.0 Project Meeting and Work Plan

When further Additional Acreage is added to the contract during the execution of the Base Period, an Extension Project Meeting and revised Work Plan will be deliverable. These are not separately deliverable for Additional Acreage included at the time of award for the Base Period.

1.1 Extension Project Meeting

Contractor will attend an Extension Project Meeting. Contractor and all principal team members will meet with NPS representatives to clarify tasks, review the approved Work Plan, and discuss any needed updates. This may be combined with a Weekly Project Meeting.

1.2 Work Plan

The Contractor will revise the approved Work Plan as necessary as additional acreage is funded and the parcel(s) and/or sub-parcels are identified by the NPS. This may require updating the project schedule. The revised Work Plan will be submitted to the NPS for review and approval within 7 calendar days.

2.0 Liming Operations

Liming Operations will be conducted in sequence with otherwise funded work, as described in the approved Work Plan.

2.1 Logistics

All logistical considerations will coincide with the Base Period effort, as described above. There should be minimal, if any, additional effort required as a result of liming additional acreage.

2.1.1 Project Area

No change from A, Base Period, 2.1.1.

2.1.2 Staging Areas

No change from A, Base Period, 2.1.2, including sub-sections. The Contractor will use the staging areas already in use and no additional sites along Skyline Drive.

2.1.3 Limestone Product

No change from A, Base Period, 2.1.3, including sub-sections.

2.2 Operations

Operations will be conducted subsequent to Base Period: Liming Operations Winter 2025-2026, as described above.

2.2.1 Dates

No change from A, Base Period, 2.2.1.

2.2.2 Limestone Application

The Contractor will lime each additional acre to the dose of the corresponding parcel or sub-parcel where it is located, as described in Table 1. The total applied limestone sands will be +5 tons/-1 ton of the total limestone sand required to lime each identified parcel or sub-parcel, measured and calculated per parcel or sub parcel.

No further change from A, Base Period, 2.2.2.

2.3 Communications

Communications for this additional activity will continue seamlessly from the effort of Base Period: Liming Operations Winter 2025-2026, as described above.

2.3.1 Weekly Project Meeting

No change from A, Base Period, 2.3.1. Weekly Project Meetings will continue through the winter between the completion of Base Period parcels and commencing Restoration and Cleanup activities.

2.3.2 Four-Day Outlook Notification

No change from A, Base Period, 2.3.2. The Contractor will make Four-Day Outlook Notifications for all Additional Acreage.

2.3.3 Status of Skyline Drive Notification

No change from A, Base Period, 2.3.3.

2.3.4 Daily Planning Brief

No change from A, Base Period, 2.3.4. The Contractor will make Daily Planning Briefs for all Additional Acreage operations.

2.3.5 Active Liming Operations Communications

No Change from A, Base Period, 2.3.5.

2.3.6 Daily Debrief

No change from A, Base Period, 2.3.6. The Contractor will make Daily Debriefs for all Additional Acreage operations.

2.3.7 Incremental Data Delivery

No change from A, Base Period, 2.3.7. The Contractor will deliver the record of liming for all Additional Acreage.

2.4 Restoration and Cleanup

Restoration and Cleanup efforts will coincide with the effort of Base Period: Liming Operations Winter 2025-2026, as described above. There should be minimal, if any, additional effort required to complete Restoration and Cleanup efforts already underway as a result of liming additional acreage. No change from A, Base Period, 2.4.

2.5 Final Inspection of Staging Areas

There will be no separate Inspection of Staging Areas. Inspections will be conducted as part of the Base Period: Liming Operations Winter 2025-2026, as described above.

3.0 Documentation and Final Report

There will be no separate Documentation Package and Final Report.

The Final Report prepared as part of the Base Period: Liming Operations Winter 2025-2026, as described above, will include content addressing all Additional Acreage limed, by parcel and/or sub-parcel where appropriate, in each of the six content areas outlined under A, Base Period, 3.0.

All documentation from liming additional acreage will be included in the Documentation Package prepared as part of the Base Period: Liming Operations Winter 2025-2026, as described above, with the following clarification:

1. Evidence of limestone acquisition will be included for all additional limestone
2. Data collected and created as part of the Treatment Verification Plan will be included for all additional acreage, referenced by parcel or sub-parcel.
3. No additional photography or videography beyond the 100 still images and 10 videos is required. The images and videos submitted as part of the Documentation Package may, however, contain content from liming the additional acreage.

Deliverables

- B1.1 Revised Work Plan(s) (format: 508 compliant Adobe .pdf)
- B2.1 Limestone Sand Application for all Additional Acreage
- B2.2 Weekly Project Meetings
- B2.3 Four-Day Outlook Notifications
- B2.4 Daily Planning Briefings
- B2.5 Daily Debriefs
- B2.6 Incremental Data Deliveries
- B3.1 Draft Final Report Inputs
- B3.2 Draft Documentation Package Content
- B3.3 Final Report Inputs
- B3.4 Final Documentation Package Content

Option Period One: Liming Operations Winter 2026-2027 (CLIN 0030)

The NPS may decide to resume liming operations the winter following the original effort. This activity is to lime 130 acres of the Project Area with a dose of 2.25 tons/acre. The NPS will determine which parcels or sub-parcels will be limed when and if this option is exercised. The requirement changes for the Option Period from the Base Period are minimal. Any such changes are detailed below.

Liming operations in Meadow Run will be in line with the Meadow Run Environmental Assessment as stipulated in the Finding of No Significant Impact. The contractor shall comply with and meet all relevant requirements under all applicable laws and regulation, including the Superintendent's Compendium, while executing the liming operations.

1.0 Project Meeting and Work Plan

1.1 Project Meeting

The Contractor will attend a project meeting within 45 days after receipt of the exercised option. All Contractor principal team members will meet virtually or in-person with NPS representatives to review the approved Work Plan from the Base Period along with the lessons learned and recommendations for process improvement from the Final Report and discuss any needed updates for the Work Plan.

An in-person Site Visit is required in conjunction with the Project Meeting if any new staging area is used.

1.2 Work Plan

The Contractor will refine their Work Plan based on the outcome of the Project Meeting and submit the Work Plan for review and acceptance by NPS representatives within 21 calendar days of completing the Project Meeting. Refinements will remain within the scope of this Statement of Work and should not alter the contract award value.

2.0 Liming Operations

Any changes from A, Base Period, are detailed below.

2.1 Logistics

The Contractor will adhere to the following logistical considerations for planning and during operations:

2.1.1 Project Area

No change from A, Base Period, 2.1.1.

2.1.2 Staging Areas

No change from A, Base Period, 2.1.2. Dates for this optional activity are below. The NPS may consider the use of other staging areas in answer to lessons learned from the Base Period. Otherwise, the Contractor should use the staging areas along Skyline Drive previously used.

2.1.2.1 Access

The Contractor will have access to the desired staging areas along Skyline Drive from 02 November 2026 through 09 April 2027 or passing final inspection, whichever is earlier. No further change from A, Base Period, 2.1.2.1.

2.1.2.2 Site Management

No change from A, Base Period, 2.1.2.2.

2.1.2.3 Vegetation

The NPS will treat the staging areas along Skyline drive before 31 October 2026. No further change from A, Base Period, 2.1.2.3.

2.1.2.4 Food Storage, Refuse, and Sanitation

No change from A, Base Period, 2.1.2.4.

2.1.3 Limestone Product

No change from A, Base Period, 2.1.3, including sub-sections.

2.2 Operations

2.2.1 Dates

Liming Operations shall be conducted during daylight hours on weekdays, Monday through Friday, from 01 December 2026 through 28 February 2027. Otherwise, no change from A, Base Period, 2.2.1.

2.2.2 Limestone Application

The Contractor will lime each parcel or sub-parcel with a dose of 2.25 tons/acre. The total applied limestone sands will be +5 tons/-1 ton of the total limestone sand required to lime each parcel or sub-parcel, measured and calculated per parcel or sub-parcel.

No further change from A, Base Period, 2.2.2.

2.3 Communications

No change from A, Base Period, 2.3, including sub-sections, unless detailed in the approved Work Plan based on lessons learned from Base Period.

2.4 Restoration and Cleanup

No change from A, Base Period, 2.4.

2.5 Final Inspection of Staging Areas

No change from A, Base Period, 2.5.

3.0 Documentation and Final Report

The Contractor shall submit a Draft Final Report and Documentation Package for review by NPS representatives within 28 calendar days of completing the Final Inspection. After NPS review,

the Contractor shall incorporate all NPS comments and edits for acceptance by NPS representatives within 14 calendar days of review.

The Final Report will include:

1. A description of the work completed
2. An analysis of the dose delivery in each parcel or sub-parcel against the method outlined in the Treatment Verification Plan
3. A comparison of the Project Schedule and target timeline to the actual completion schedule
4. Obstacles and risks encountered during the work and how they were managed
5. Any safety issues and incidents and how they were addressed
6. Results of any attempted process improvements
7. Recommendations for further process improvement for future similar projects

The Documentation Package will include:

1. Evidence of limestone acquisition; this may include copies of all invoices and load slips from liming material
2. Data collected and created as part of the Treatment Verification Plan. This includes all equipment instrumentation and geospatial data, as well as a clean library containing all data included in the Incremental Data Delivery. All geospatial data will be in a commonly used file type, i.e. Esri shapefile or Google kml/kmz.
4. Still photography and videography of liming operation, including: 100 still images and 10 videos, each 2-10 minutes in length; a minimum of 25% of the content should be from the perspective of the application crew(s), a novel perspective of the application, and/or close up of their same work. Videography should be captured at 4K resolution or higher. Photography should be shot in RAW format at the highest resolution any camera allows.

Deliverables

- C1.1 Project Meeting
- C1.2 One (1) Work Plan (format: 508 compliant Adobe .pdf)
- C2.1 Limestone Sand Application for each Parcel and/or Sub-Parcel
- C2.2 Weekly Project Meetings
- C2.3 Four-Day Outlook Notifications
- C2.4 Daily Planning Briefings
- C2.5 Daily Debriefs
- C2.6 Incremental Data Deliveries
- C2.7 Final Inspection of Staging Areas
- C3.1 One (1) Draft Final Report (format: MS Word)
- C3.2 One (1) Draft Documentation Package
- C3.3 One (1) Final Report (format: 508 compliant Adobe .pdf)
- C3.4 One (1) Final Documentation Package

C. Increase Quantity Option: Liming Additional Acreage Winter 2026-2027 (CLIN 0040)

The NPS may decide to lime additional acreage during the Option Period: Liming Operations Winter 2026-2027, as described above. This activity is to lime up to an additional not to exceed 1,460 acres of the Project Area. The NPS will identify which parcels or sub-parcels (if an entire parcel is not funded) will be limed. Each acre will be limed to the dose of the corresponding parcel where it is located, as described in Table 1.

Liming Additional Acreage for this Option will be completed in conjunction with the effort of Option Period One: Liming Operations Winter 2026-2027, as described above. Liming Operations will be conducted in sequence with otherwise funded work and follow the same requirements as detailed above, respectively, and as described in the approved Work Plan. Inputs will be provided for the corresponding Final Report and Documentation Package. Specific changes to any requirements are below.

1.0 Project Meeting and Work Plan

When further Additional Acreage is added to the contract during the execution of the Option Period, an Extension Project Meeting and revised Work Plan will be deliverable. These are not separately deliverable for Additional Acreage included at the time of exercising the Option Period.

1.1 Extension Project Meeting

Contractor will attend an Extension Project Meeting. Contractor and all principal team members will meet with NPS representatives to clarify tasks, review the approved Work Plan, and discuss any needed updates. This may be combined with a Weekly Project Meeting.

1.2 Work Plan

The Contractor will revise the approved Work Plan as necessary as additional acreage is funded and the parcel(s) and/or sub-parcels are identified by the NPS. This may require updating the project schedule. The revised Work Plan will be submitted to the NPS for review and approval within 7 calendar days.

2.0 Liming Operations

Liming Operations will be conducted in sequence with otherwise funded work, as described in the approved Work Plan.

2.1 Logistics

All logistical considerations will coincide with the effort of the Optional Period: Liming Operations Winter 2026-2027, as described above. There should be minimal, if any, additional effort required as a result of liming additional acreage.

2.1.1 Project Area

No change from A, Base Period , 2.1.1.

2.1.2 Staging Areas

No change from C, Option Period , 2.1.2, including sub-sections. The Contractor will use the staging areas already in use and no additional sites along Skyline Drive.

2.1.3 Limestone Product

No change from A, Base Period , 2.1.3, including sub-sections.

2.2 Operations

Operations will be conducted subsequent to Option Period effort, as described above.

2.2.1 Dates

No change from C, Option Period, 2.2.1, respectively.

2.2.2 Limestone Application

The Contractor will lime each additional acre to the dose of the corresponding parcel or sub-parcel where it is located, as described in Table 1. The total applied limestone sands will be +5 tons/-1 ton of the total limestone sand required to lime each identified parcel or sub-parcel, measured and calculated per parcel or sub parcel.

No further change from A, Base Period , 2.2.2.

2.3 Communications

Communications for this additional activity will continue seamlessly from the effort of the Option Period: Liming Operations Winter 2026-2027, as described above.

2.3.1 Weekly Project Meeting

No change from A, Base Year, 2.3.1. Weekly Project Meetings will continue through the winter between the completion of Option Period parcels and commencing Restoration and Cleanup activities.

2.3.2 Four-Day Outlook Notification

No change from A, Base Period , 2.3.2. The Contractor will make Four-Day Outlook Notifications for all Additional Acreage.

2.3.3 Status of Skyline Drive Notification

No change from A, Base Period , 2.3.3.

2.3.4 Daily Planning Brief

No change from A, Base Period, 2.3.4. The Contractor will make Daily Planning Briefs for all Additional Acreage operations.

2.3.5 Active Liming Operations Communications

No Change from A, Base Period, 2.3.5.

2.3.6 Daily Debrief

No change from A, Base Period, 2.3.6. The Contractor will make Daily Debriefs for all Additional Acreage operations.

2.3.7 Incremental Data Delivery

No change from A, Base Period, 2.3.7. The Contractor will deliver the record of liming for all Additional Acreage.

2.4 Restoration and Cleanup

Restoration and Cleanup efforts will coincide with the effort of the Option Period: Liming Operations Winter 2026-2027, as described above. There should be minimal, if any, additional effort required to complete Restoration and Cleanup efforts already underway as a result of liming additional acreage. No change from A, Base Period, 2.4.

2.5 Final Inspection of Staging Areas

There will be no separate Inspection of Staging Areas. Inspections will be conducted as part of the Option Period: Liming Operations Winter 2026-2027, as described above.

3.0 Documentation and Final Report

There will be no separate Documentation Package and Final Report.

The Final Report prepared as part of the Option Period: Liming Operations Winter 2026-2027, as described above, will include content addressing all Additional Acreage limed, by parcel and/or sub-parcel where appropriate, in each of the six content areas outlined under A, Base Period, 3.0.

All documentation from liming additional acreage will be included in the Documentation Package prepared as part of the Option Period: Liming Operations Winter 2026-2027, as described above, with the following clarification:

1. Evidence of limestone acquisition will be included for all additional limestone
2. Data collected and created as part of the Treatment Verification Plan will be included for all additional acreage, referenced by parcel or sub-parcel.
3. No additional photography or videography beyond the 100 still images and 10 videos is required. The images and videos submitted as part of the Documentation Package may, however, contain content from liming the additional acreage.

Deliverables

- D1.1 Revised Work Plan(s) (format: 508 compliant Adobe .pdf)
- D2.1 Limestone Sand Application for all Additional Acreage
- D2.2 Weekly Project Meetings
- D2.3 Four-Day Outlook Notifications
- D2.4 Daily Planning Briefings
- D2.5 Daily Debriefs

- D2.6 Incremental Data Deliveries
- D3.1 Draft Final Report Inputs
- D3.2 Draft Documentation Package Content
- D3.3 Final Report Inputs
- D3.4 Final Documentation Package Content